Auctioning (EU ETS 1): German Auctioning of Emission Allowances

Annual Report 2024



EU ETS 1

Umwelt Bundesamt

DEHSt
Deutsche Emissionshandelsstelle

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Status: March 2025

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German Auctioning of Emissions Allowances

The German auctions of the fourth trading period are held weekly on the European Energy Exchange (EEX) spot market. The German Emissions Trading Authority (DEHSt) at the German Environment Agency is the competent authority for the German auctions. Further updated information on the fourth trading period auctions can be downloaded from the EEX website. Reports on the second, third and fourth trading periods auction results are available on the DEHSt auction website. The 2024 and 2025 auction calendars are published on the EEX website.

The following tables summarise the key data of the German auctions on the EEX in 2024 and 2025:

2024		
Auction product	**Fourth trading period** EUA spot contracts	**Fourth trading period** EUAA spot contracts
Period	19/01/2024 to 13/12/2024	16/10/2024
Auction volume entire year	83,991,500 EUA[1]	1,061,000 EUAA
Auction volume per auction	Jan-Aug[2]: 1,796,000 EUA Sep-Dec: 1,886,500 EUA	1,061,000 EUAA on 16/10/2024
Auction frequency	Weekly (Fridays, 09:00 to 11:00 am)	On 16/10/2024 (09:00 to 11:00 am)
Bid volume	500 EUA/EUAA	
Auction mode	Single-round, uniform-price procedure with a closed order book	

2025	
Auction product	Fourth trading period **EUA spot contracts**[3]
Period	10/01/2025 to 12/12/2025
Auction volume entire year	96,764,500 EUA (provisional)[4]
Auction volume per auction	Jan-Aug: 1,607,000 EUA Sep-Dec: 3,129,500 EUA (provisional)
Auction frequency	Weekly (Fridays, 09:00 to 11:00 am)
Bid volume	500 EUA
Auction mode	Single-round, uniform-price procedure with a closed order book

1 The MSR reduction for the auction volumes from September 2024 to August 2025 was published by the European Commission on 1st June 2024

2 From 2024 onwards, the volumes to be auctioned in August will no longer be reduced by half as in the past. This is in line with the requirements of the new EU Auctioning Regulation.

3 From 2025 onwards, aviation allowances (EUAA) will no longer be auctioned on the primary market. The corresponding volumes will be auctioned proportionally as EUA spot contracts in the weekly auctions. This is in line with the requirements of the new EU Auctioning Regulation.

4 The MSR reduction for the auction volumes from September 2025 to August 2026 is expected to be determined by the European Commission in June 2025.

1 Overview

1.1 Overview of the entire year 2024

In 2024 a total of about **85 million EUA and EUAA** (2023: 91 million) with a total value of about **€ 5.5 billion** (2023: € 7.7 billion) were auctioned at EEX for Germany. This resulted in a volume-weighted average price per allowance of **€ 65** (2023: € 83,66).

The total annual revenue was about 28 percent below the record level of the previous year. This is mainly due to the significantly lower price level on the carbon market as well as lower auction volumes. At the EUA auction on 23rd February, the lowest price since May 2021 was achieved at €49.50 (see also Figure 1).

The following table summarizes the auction results on a monthly basis differentiated by auctions for EUA and EUAA.

Table 1: Overview of the entire year 2024

Month	Type	Auction volume	Bid volume	Cover ratio	Bidders	Successful bidders	Price	Revenue
January	EUA	3,592,000	7,586,000	*2.11	*26	*9	*€ 61.94	222,488,480 €
February	EUA	7,184,000	12,874,000	*1.79	*23	*21	*€ 55.70	400,112,880 €
March	EUA	7,184,000	14,021,000	*1.95	*24	*17	*€ 56.60	406,578,480 €
April	EUA	7,184,000	14,477,000	*2.02	*27	*17	*€ 64.65	464,463,560 €
May	EUA	7,184,000	14,492,000	*2.02	*24	*14	*€ 71.98	517,122,280 €
June	EUA	7,184,000	13,068,000	*1.82	*23	*18	*€ 67.67	486,141,280 €
July	EUA	7,184,000	13,844,000	*1.93	*24	*18	*€ 66.70	479,172,800 €
August	EUA	8,994,000	18,068,500	**2.01	*22	*16	**€ 70.40	633,208,900 €
September	EUA	7,546,000	14,641,000	*1.94	*24	*16	*€ 65.13	491,470,980 €
October	EUA	7,546,000	14,893,000	*1.97	*22	*17	*€ 63.24	477,209,040 €
	EUAA	*1,061,000*	*3,212,000*	*3.03*	*15*	*6*	*€ 64.14*	*68,052,540 €*
November	EUA	9,432,500	18,342,000	*1.94	*22	*19	*€ 66.89	630,939,925 €
December	EUA	3,777,000	7,194,500	**1.91	24	20	**€ 66.66	251,772,460 €
EUA		**83,991,500**	**163,501,000**	****1.95**	***24**	***17**	****€ 65.01**	**5,460,681,065 €**
EUAA		***1,061,000***	***3,212,000***	***3.03***	***15***	***6***	***€ 64.14***	***68,052,540 €***
Total		**85,052,500**	**166,713,000**	****1.96**	***23**	***17**	****€ 65.00**	**€ 5,528,733,605**

Source: EEX, DEHSt
* average
** volume-weighted average

As in previous years, the deviations of the EUA auction prices from the reference prices on the lead markets were mostly in the tenths of a percent range. The average deviation just before the end of the bidding period (11:00 am) was 0.36 percent or € 0.23 (see also Section 3.4). A major factor behind this development was the broadly stable demand on the primary market.

Overall, valid bids totaling about 164 million allowances were entered in the trading system across the 46 EUA-auction dates. Thus, the average cover ratio in the reporting period was 1.95 (see also Section 2.2). In 2024 the average number of participants in the German EUA auctions was 24, the average number of successful bidders amounted to 17. Bidder participation in 2024 was thus higher than in the previous year. From a bidder's perspective, the average success rate over all auction dates was about 73 percent (see also Section 2.1).

In addition, 1,061,000 aviation allowances (EUAA) with a total value of about € 68 million were auctioned for Germany on 16th October. The auction price for the 6 successful bidders was € 64.14. Overall, 15 bidders with valid bids totalling about 3.2 million allowances participated in the auction. The auction on 16th October was the last EUAA auction for Germany. From 2025 onwards, EUAA will no longer be offered on the primary market. In future, the corresponding volumes will be auctioned on a proportional basis as EUA spot contracts at the weekly auctions. This is in line with the requirements of the new EU Auctioning Regulation.

Overall, last year's auctions were implemented in accordance with the applicable national and European regulations (see also Section 4).

1.2 Overview since the beginning of the third trading period in 2013

The following table summarizes the auction results since the beginning of the third trading period (2013–2020) on a yearly basis. Although the general regulatory framework of the EU Auctioning Regulation has remained largely unchanged since the start of the third trading period and also applies in principle to the fourth trading period (2021–2030), major decisions were taken during this period to determine the amount of auction volumes. In addition, this period was characterized by significant changes in the European carbon market, which had a considerable impact on price levels.

Between 2014 and 2016 the development of the auction volumes was substantially affected by the so-called "backloading"-decision. Accordingly, 900 million emission allowances less were auctioned EU-wide than originally planned.[5] In line with this decision, German auction volumes were reduced by some 175 million EUA over the same period. In addition, the Market Stability Reserve (MSR) came into effect at the beginning of 2019: the auction volume was reduced EU-wide by over 2 billion EUA in the period 2019 to 2024 in accordance with the MSR mechanism. The German auction volumes were reduced proportionately by almost 500 million EUA in this period.

In addition to these auction volume adjustments, the EUA-price development on the lead markets had a major effect on the revenue development of the individual years. In 2013, the average revenue of € 4.33 per allowance was significantly below the previous year's figure (referred to as so-called "early auctions"[6]). In April and May 2013, the so far lowest auction price at a German EUA auction was reached at € 2.95 (see also Figure 1). In the period from 2014 to 2017, the average annual auction prices ranged between about € 5.20 and € 7.60. As a result of a strong upward trend on the carbon market, average revenues of almost € 15 were achieved in 2018 and almost € 25 in in each of the following years. In 2021, the average price had already exceeded the € 50 mark and was above 80 euros in both 2022 and 2023. In 2024, the average auction price of € 65 was around 22 percent below the previous year's figure. On an annual basis, this is the first time since 2016 that the average auction price has fallen significantly (see also Figure 2).

5 The auction volumes were reduced as follows: 400 million EUA in 2014, 300 million EUA in 2015 and 200 million EUA in 2016. The overall backloading volume of 900 million EUA was transferred to the Marker Stability Reserve (MSR) in 2019.

6 In the Fourth Quarter of 2012, some 90 million EUA of the third trading period were auctioned in the EU. These amounts have been deducted proportionately from the planned auction volumes of 2013 and 2014.

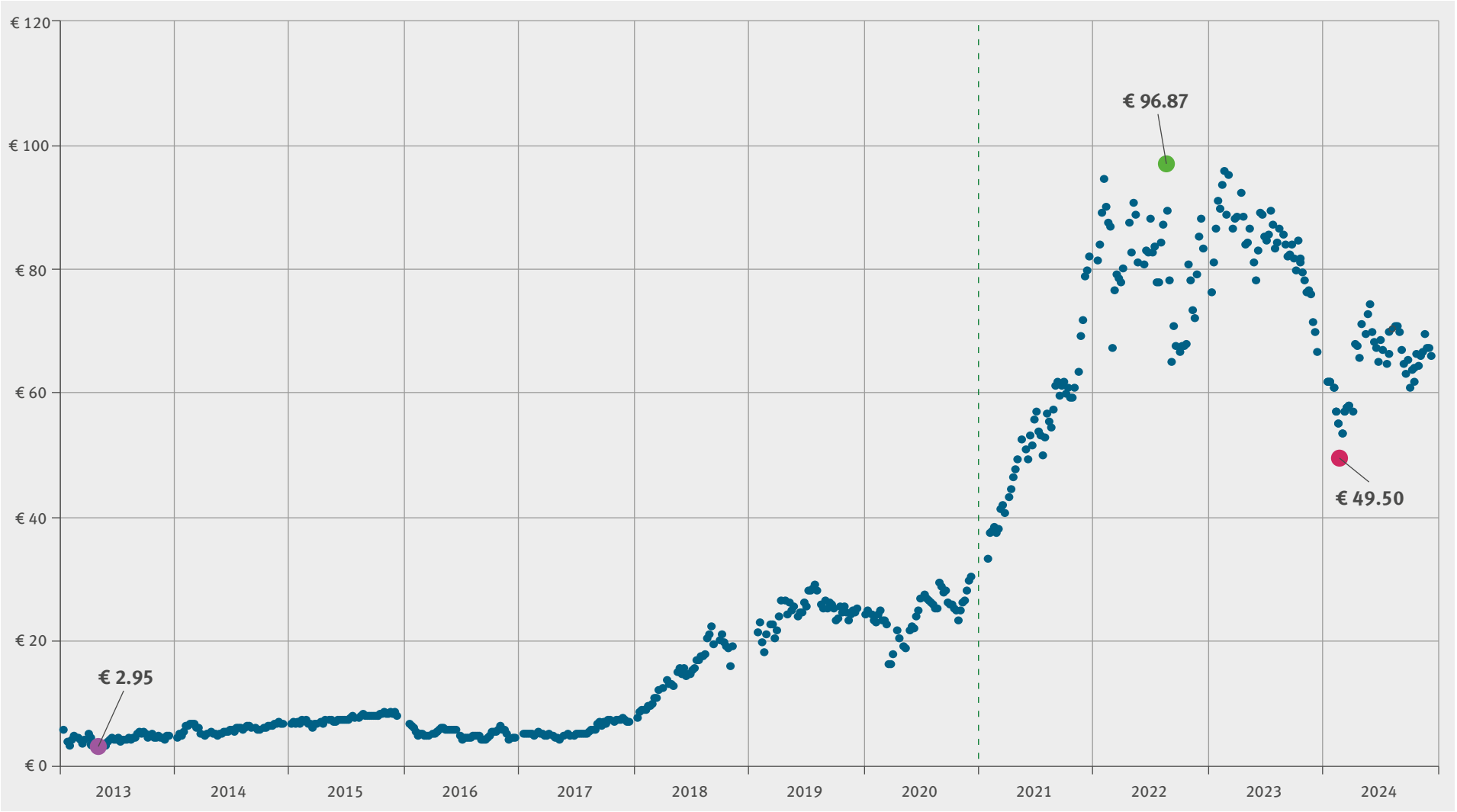
The auctions of aviation allowances (EUAA) were significantly influenced by the adjustment of the scope of the Emissions Trading Directive in relation to aviation emissions. Firstly, the scope has been substantially reduced by the EU's so-called "stop-the-clock" decision in 2012. Furthermore, an additional restriction was made for the period from 2013 to 2020 ("reduced scope"). As a result, EU-wide EUAA-auction volumes from 2012 onwards were noticeably lower than originally planned. In line with the European requirements, Germany carried out a total of twelve EUAA auctions in the years 2012 and 2015 to 2024. The auction on 16th October 2024 was the last auction of EUAA for Germany (see also Chapter 1.1).

Table 2: Overview since the start of the third trading period

Year	Type	Auction volume	Bid volume	Cover ratio	Bidders	Successful bidders	Price	Revenue
2012	EUA	23,531,000	86,799,000	**3.69	*14	*7	**€ 7.06	€ 166,178,670
	EUAA	*2,500,000*	*4,552,000*	*1.82*	*6*	*5*	*€ 7.01*	*€ 17,525,000*
2013	EUA	182,560,500	529,511,500	**2.90	*18	*14	**€ 4.33	€ 791,253,420
2014	EUA	127,127,500	614,996,000	**4.84	*16	*11	**€ 5.90	€ 749,973,880
2015	EUA	143,893,500	412,946,500	**2.87	*17	*13	**€ 7.60	€ 1,093,312,500
	EUAA	*2,229,000*	*7,665,500*	***3.44*	**12*	**6*	***€ 7.57*	*€ 16,867,680*
2016	EUA	160,770,500	358,232,500	**2.23	*18	*13	**€ 5.26	€ 845,739,870
	EUAA	*857,500*	*3,362,500*	*3.92*	*10*	*6*	*€ 5.42*	*€ 4,647,650*
2017	EUA	196,820,000	493,172,500	**2.51	*20	*15	**€ 5.80	€ 1,141,744,975
	EUAA	*675,500*	*2,525,000*	*3.74*	*7*	*6*	*€ 7.51*	*€ 5,073,005*
2018	EUA	172,220,000	436,379,500	**2.53	*24	*18	**€ 14.90	€ 2,565,336,800
	EUAA	*800,500*	*1,823,500*	*2.28*	*9*	*6*	*€ 20.38*	*€ 16,314,190*
2019	EUA	127,561,500	234,026,000	**1.83	*22	*17	**€ 24.66	€ 3,146,135,125
	EUAA	*801,500*	*2,377,000*	*2.97*	*9*	*5*	*€ 22.32*	*€ 17,889,480*
2020	EUA	107,433,000	209,032,500	**1.95	*23	*16	**€ 24.59	€ 2,641,787,980
	EUAA	*769,000*	*2,307,000*	*3.00*	*10*	*5*	**€ 26.76*	*€ 20,578,440*
2021	EUA	100,462,500	171,888,000	**1.71	*23	*17	**€ 52.47	€ 5,270,913,940
	EUAA	*600,000*	*1,785,000*	*2.98*	*8*	*4*	*€ 58.75*	*€ 35,250,000*
2022	EUA	84,230,000	185,569,500	**2.20	*19	*14	**€ 80.40	€ 6,772,399,130
	EUAA	*586,500*	*1,178,000*	*2.01*	*6*	*4*	*€ 68.62*	*€ 40,245,630*
2023	EUA	90,609,000	193,077,500	**2.13	*20	*14	*€ 83.68	€ 7,582,509,030
	EUAA	*907,000*	*1,950,500*	*2.15*	*13*	*12*	*€ 81.72*	*€ 74,120,040*
2024	EUA	83,991,500	163,501,000	**1.95	*24	*17	**€ 65.01	€ 5,460,681,065
	EUAA	*1,061,000*	*3,212,000*	*3.03*	*15*	*6*	*€ 64.14*	*€ 68,052,540*
EUA		**1,601,210,500**	**4,089,132,000**	****2.55**	**20**	**14**	****€ 23.87**	**€ 38,227,966,385**
EUAA		***11,787,500***	***32,738,000***	*****2.78***	****10***	****6***	*****€ 26.86***	***€ 316,563,655***
Gesamt		**1,612,998,000**	**4,121,870,000**	****2.56**	**20**	**14**	****€ 23.90**	**€ 38,544,530,040**

Source: EEX, DEHSt
* Average
** Volume-weighted averaget

The following chart shows the auction prices of the German EUA auctions held since 2013 (dots). The lowest (purple dot) and the highest (green dot) auction price levels achieved during this period, as well as the lowest price in 2024 (red dot, see also section 1.1) are indicated separately.

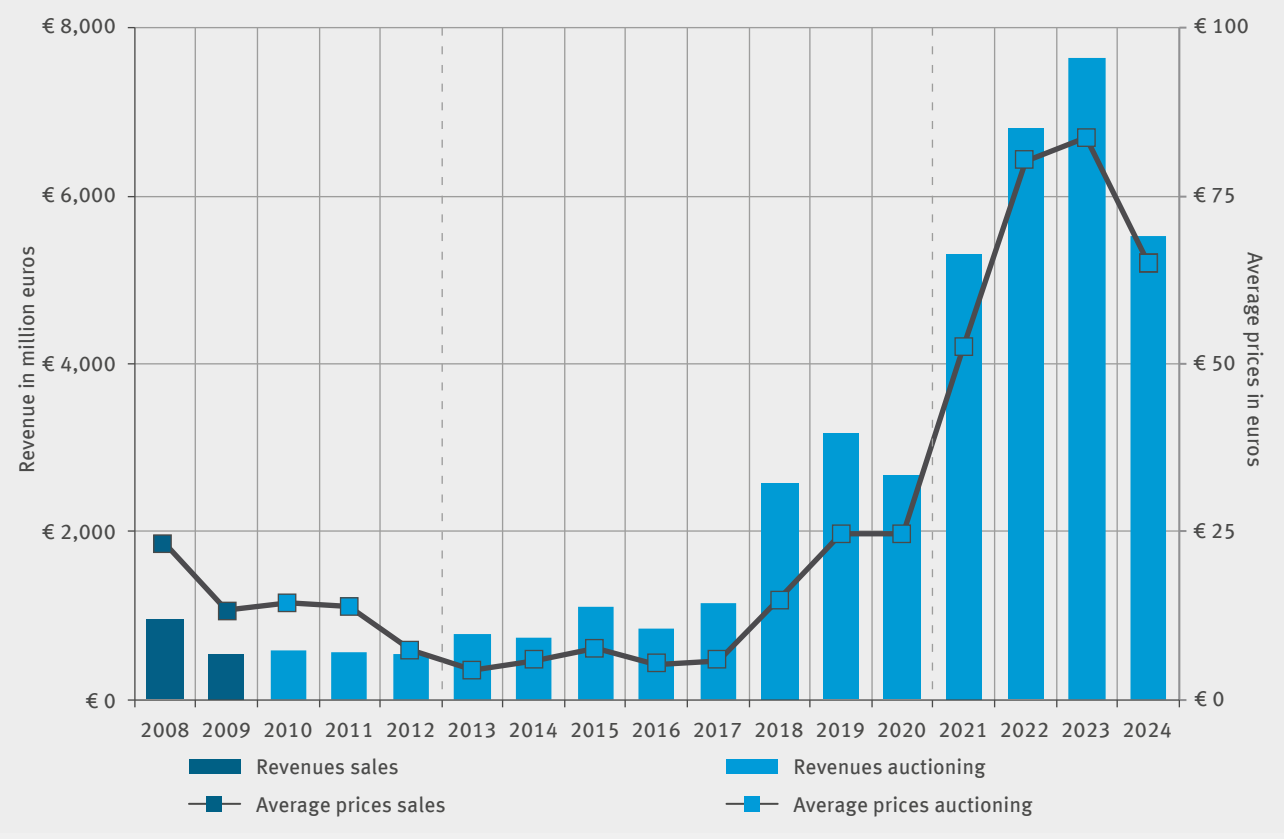


Source: EEX, DEHSt

Figure 1: German auction prices since 2013

The following chart shows the annual revenues from the sales (2008-2009) and auctioning (since 2010) of emission allowances for Germany and the average prices achieved. In addition to the auction volumes of the third and fourth trading period, the figure thus also includes the sales and auction volumes of the second trading period (2008-2012). In 2023, revenues reached a record level and exceeded the 7 billion euro mark for the first time, before falling to below 6 billion in 2024.

Between 2008 and 2024, the Federal Government raised a total of more than **€ 41 billion** from the sale of over **1.8 billion emissions allowances**. In Germany, all auction revenues have been flowing into the Climate and Transformation Fund (KTF)[7] since 2012. Already before 2012, a large proportion of the sales revenues had already been channeled into national and international projects of the German government. Since January 2010, a total of over **800 auctions** have been successfully held for Germany on the EEX.



Source: EEX, DEHSt

Figure 2: Revenues from sales/auctioning and average prices since 2008

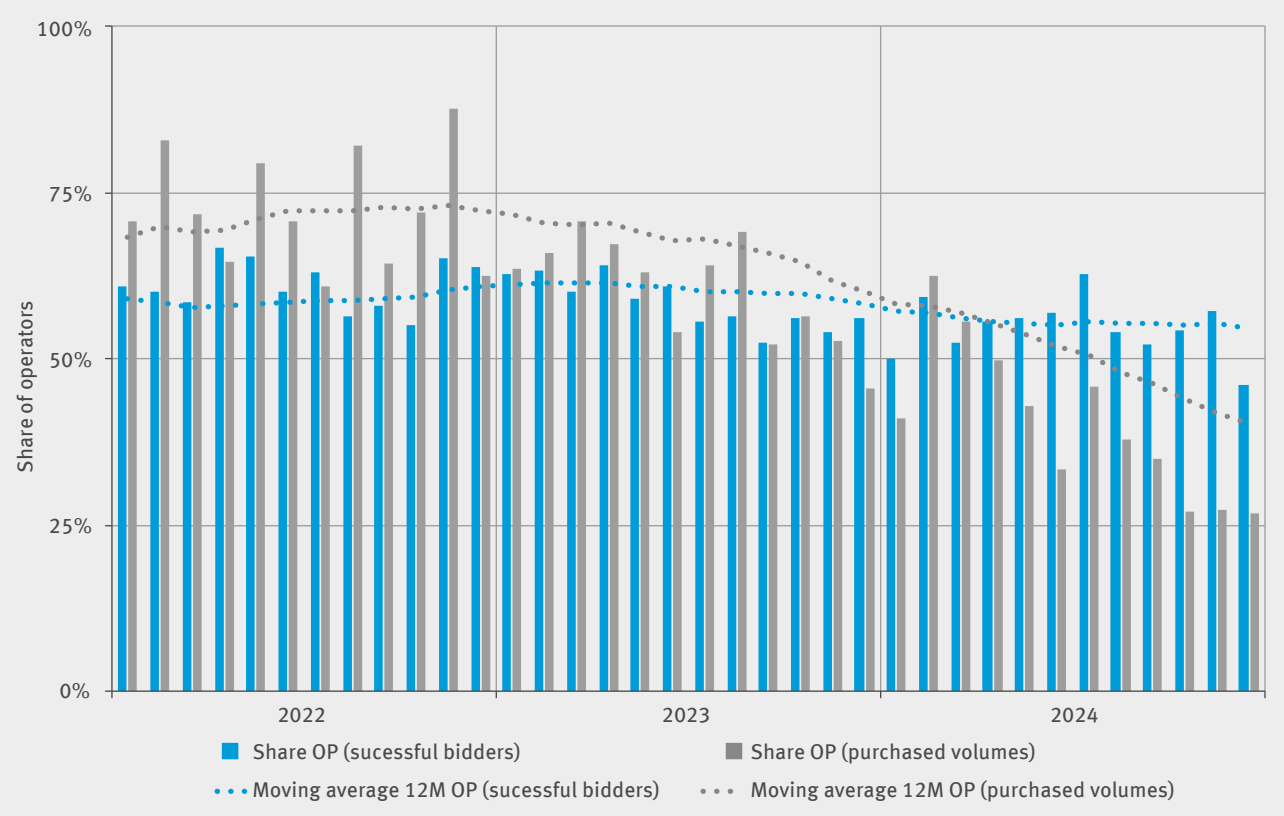
7 Until July 2022 – Energy and Climate Fund (EKF).

2 Bidder Participation

2.1 Participation

By the end of December 2024, 111 trading members of EEX had been authorized to submit bids in the primary market auctions (December 2023: 100). In 2024 a total of 33 different bidders (2023: 37) were successful in the German primary market auctions, including 20 operators (i. e. operators of ETS covered installations or aircraft operators) as well as 13 non-compliance companies (2023: 23/14). The 20 operators purchased a total of approximately 40 percent of the auctioned volumes (2023: 60 percent).

The following figure illustrates the development of the proportion of operators among the successful bidders as well as of the purchased auction volumes since 2022. Moving averages over the last 12 months for the share of auctioned volumes purchased show a declining trend from September 2023 onwards, with a significant acceleration in the course of 2024. Overall, operators purchased approximately 27 percent of the auctioned volumes in December 2024. The share purchased by operators in the weekly auctions was thus below 50 percent for the ninth consecutive month and at a similar level to the previous two months, the lowest level since August 2016. The share of successful bidders, on the other hand, was relatively stable and close to the moving average of the last 12 months.



Source: EEX, DEHSt
OP = Operators

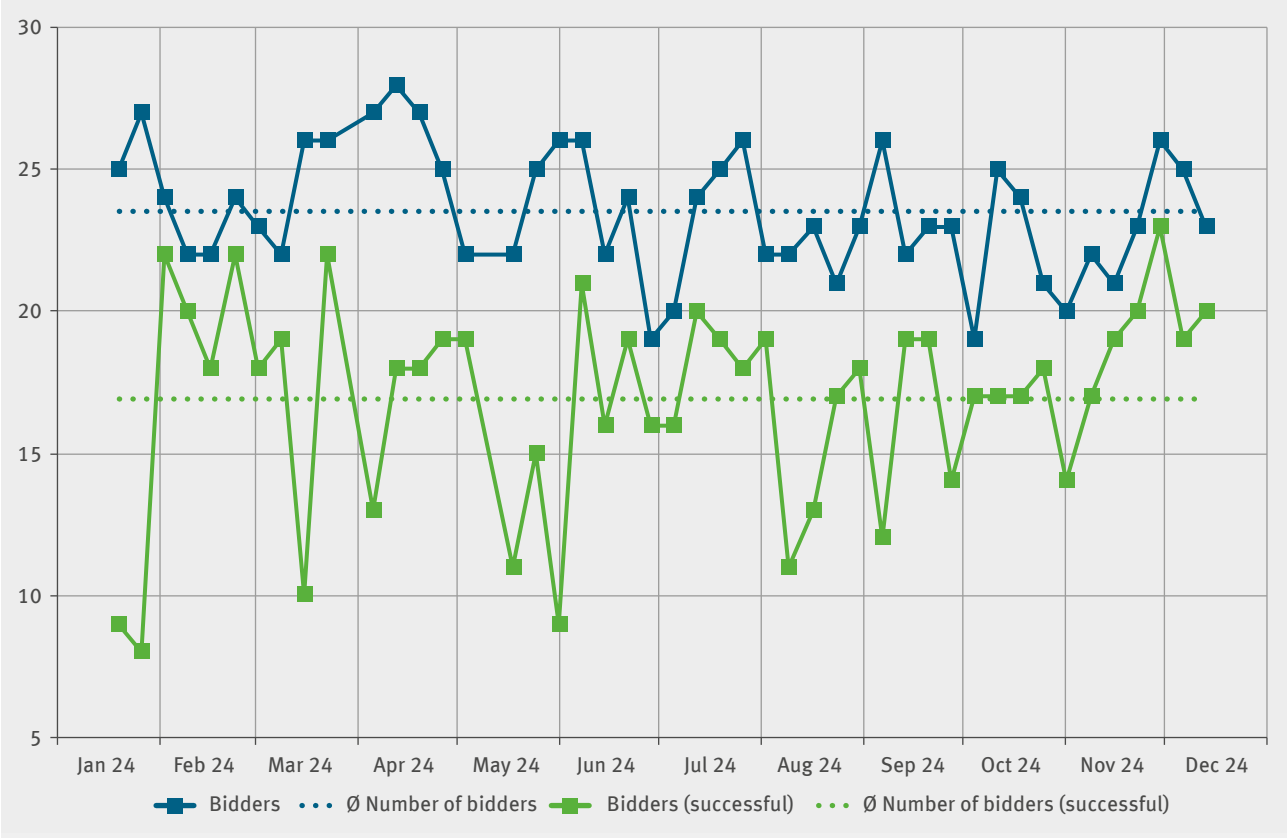
Figure 3: Share of operators among successful bidders and purchased auction volumes since 2022 (EUA)

Participation in the **EUA auctions** fluctuated during the reported period between 19 and 28 bidders (2023: 15 to 24). Thus, all approved bidders together were never simultaneously active in one auction. The number of successful participants varied between 8 and 23 bidders (2023: 6 to 21). There was no such auction where all participating bidders were successful.

The number of participants in the German **EUA auctions** in 2024 averaged across all dates was around 24 bidders (2023: 20). The average number of successful bidders averaged around 17 (2023: 14). The average ratio of successful to participating bidders was thus around 73 percent (2023: 71 percent). Therefore, on an annual average, about 3 out of 4 auction participants were awarded allowances in a German EUA auction. At almost all auction dates, the ratio of successful to participating bidders was above a level of 50 percent. The probability of successfully being awarded emission allowances in an auction was therefore relatively high from the average bidder's perspective.

With 15 bidders (2023: 13), the participation in the **aviation allowances** auction in October was below the average participation in the EUA auctions.

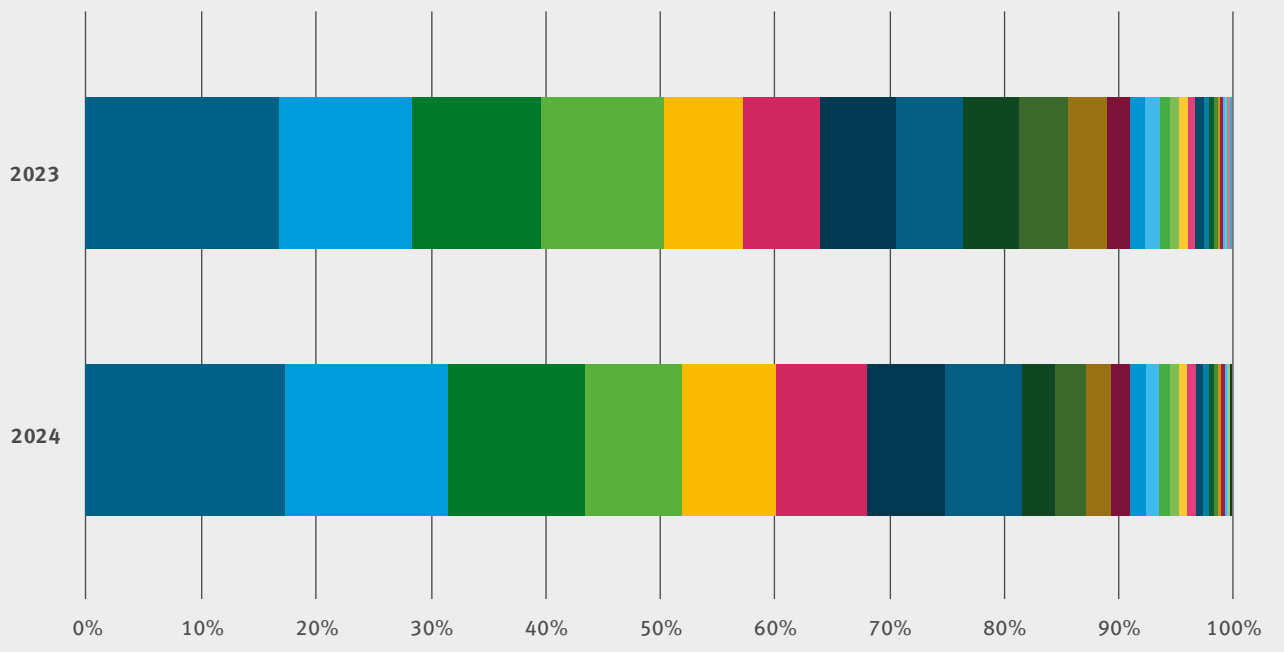
The following figure illustrates EUA auction participation over time for the year 2024.



Source: EEX, DEHSt

Figure 4: Number of participating and successful bidders per EUA auction date

The following chart shows the shares of the individual bidders in the aggregated purchased volumes in 2023 and 2024. The colours are based exclusively on the order ranked by purchased volumes and are not linked to the individual bidders. In 2024, the five largest bidders by volume together accounted for just over 60 percent of the annual auction volume, a slight increase in the share of this top group compared to the previous year.



Source: EEX, DEHSt

Figure 5: Share of successful bidders in purchased volume in 2023 and 2024

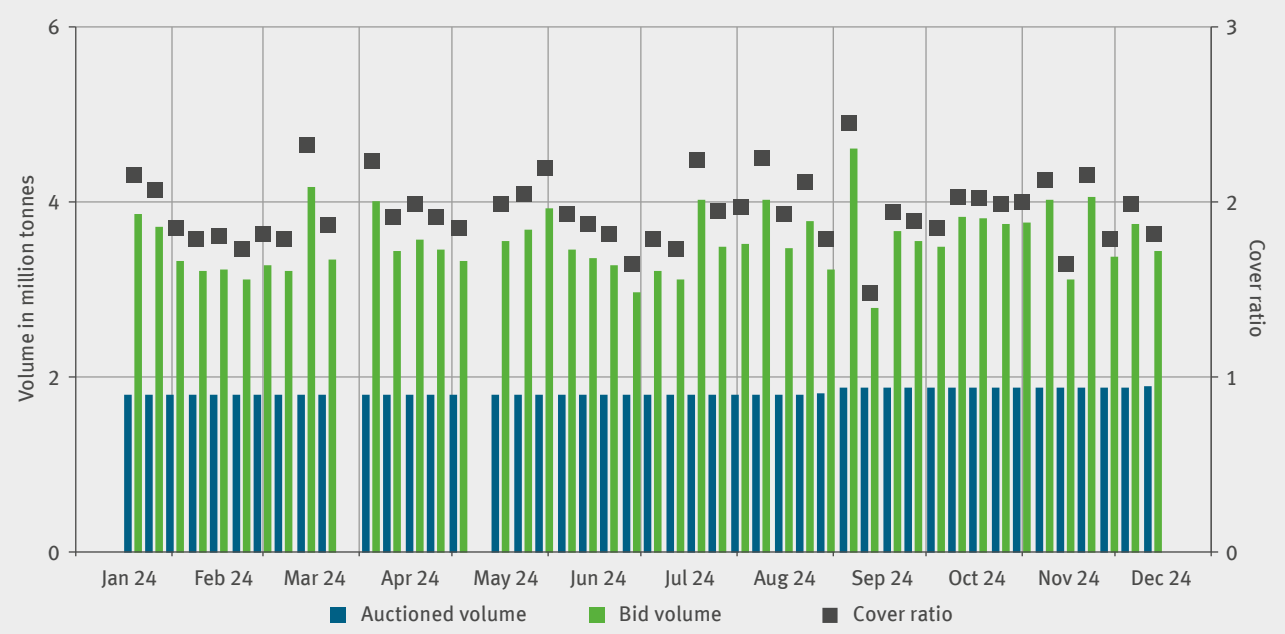
2.2 Volume Demand

In 2024, a total of valid bids amounting to some 164 million allowances were made in the German EUA auctions (2023: 193 million). In relation to the total auctioned volume, this determines an average cover ratio of 1.95 (2023: 2.13). Deviations of the clearing prices from the reference prices on the lead markets remain marginal and were mostly in the tenths of a percent range (see also Section 3.3). The demand situation on the primary market can therefore be described as robust.

The aggregated bid volumes for each EUA auction date fluctuated during the reporting period from 2.8 to 4.6 million allowances (2023: 1.9 to 5.2 million). The resulting cover ratio for each auction date varied within a range of 1.48 to 2.45 (2023: 1.58 to 3.78). It should be noted that supply volumes in the individual auctions varied to some extent. Since the beginning of 2019, the Market Stability Reserve (MSR) has come into effect. Hence, the 2024 EUA auction volume was reduced by the MSR scheme. Between January and August, about 1,8 million EUA were offered per auction date. From 2024, the volumes to be auctioned in August are no longer reduced by half as before in accordance with European regulatory requirements. Between September and December, about 1.9 million EUA were auctioned per date.

The cover ratio in the **EUAA auction** in October was 3.03 (2023: 2.15) and thus within the fluctuation range of the cover ratios in EUA auctions.

The following figure summarizes the main findings for each EUA auction date.



Source: EEX, DEHSt

Figure 6: Auctioned and bid volume and cover ratio for each EUA auction date

The average bid volume per auction participant in an EUA auction, averaged over all dates, was approximately 150,000 allowances (2023: 205,000 EUA). The average award volume was around 110,000 allowances per successful bidder (2023: 135,000 EUA).

In the EUAA auction the bid volume per auction participant was about 215,000 allowances (2023: 135,000 EUAA). The average volume awarded to each successful bidder was about 175,000 allowances (2023: 75,000 EUAA).

Awarding allowances at the auctions is based on the EU-wide applicable uniform price procedure with one prescribed round of bidding (also see Info box below).

Uniform Price Auction at the EEX

The auctions at the EEX follow the uniform price procedure which is common practice throughout the European Union. It involves one round of bidding and a closed order book. This simple and robust procedure protects against market manipulation and was used already for German auctions during the second trading period. The uniform price procedure means that all successful bidders pay the same price. The auction clearing price is determined by ranking all eligible bids, beginning with the highest bid, according to the bidding price.

In the event of equal bids, the bids are sorted by a random process. The bid volumes are aggregated starting with the highest bid until the volume of emission allowances on offer (e. g. 1.9 million EUA) is reached. The bid price at which the aggregate volume of bids reaches or exceeds the offered volume of auctioned allowances determines the auction clearing price.

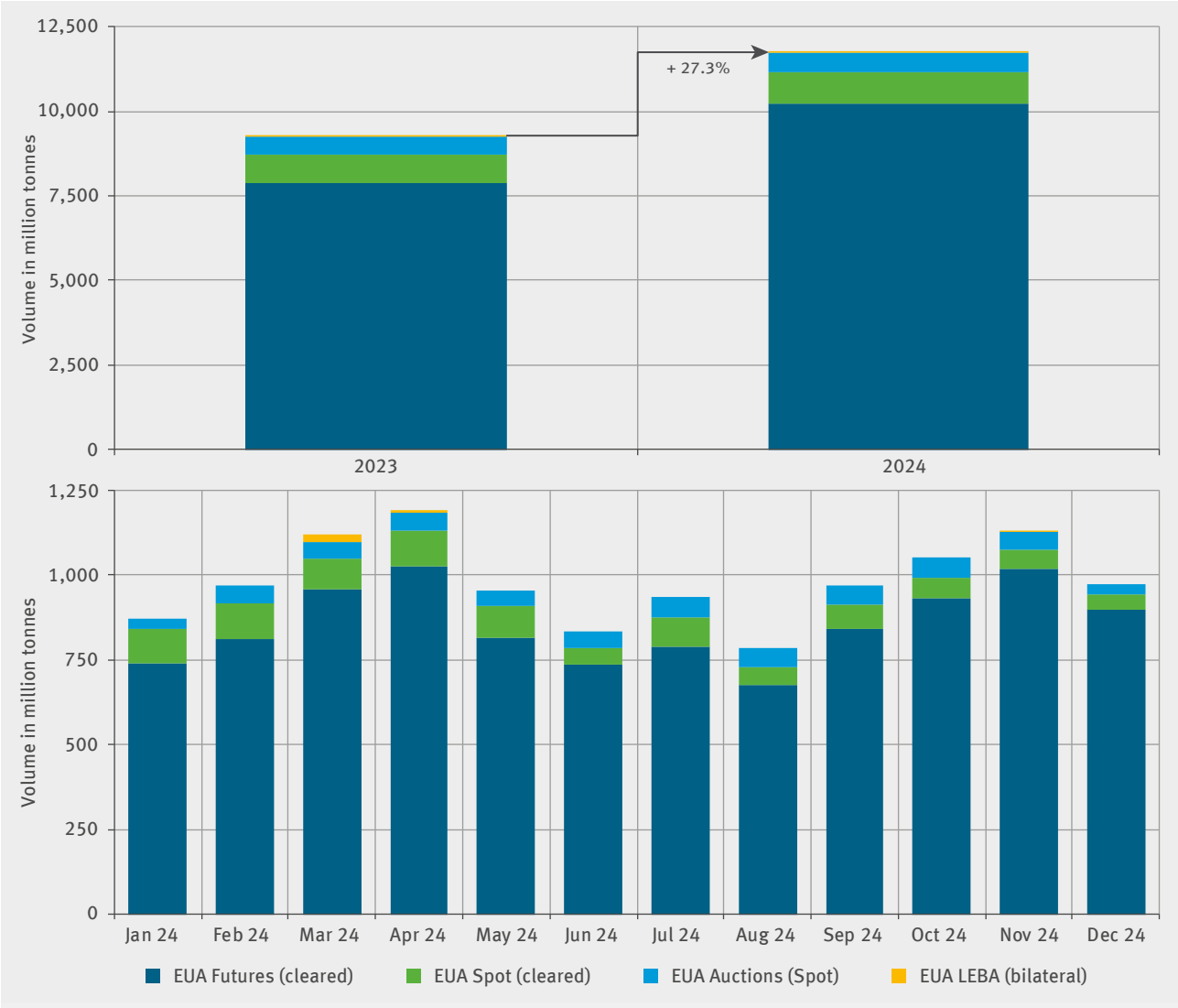
Bidders will thus be awarded their bid volume if their bidding price at least matches the auction clearing price. If a bid matches the clearing price exactly, the bidder may be awarded the remaining allowances which may be below the volume requested, depending on the bidding situation. If the bid from several participants matches the auction clearing price exactly, the above-mentioned random process applies.

3 Auctions and the Secondary Market

3.1 Trading Volume EUA in 2024

In 2024, a total of roughly 84 million EUA were auctioned at EEX for Germany (primary market). During the reporting period, the cumulative trading volume in the relevant total market[8] was about **11.8 billion EUA** (2023: 9.3 billion). This comprises the monthly data of the London Energy Brokers' Association (LEBA).[9] The trading volume on the **secondary market** increased by almost 30 percent in 2024 compared to the previous year, but did not quite reach the record level of 2021. Due to the overall lower EUA price level, the turnover of more than € 700 billion was in a similar range to the previous year.

The figure below illustrates the monthly trading volumes separately for different market segments.



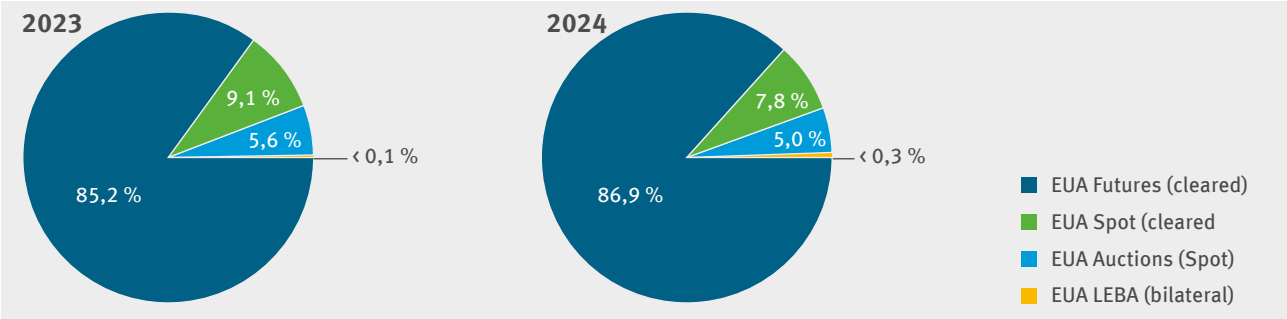
Source: ICE, EEX, Nasdaq OMX, Refinitiv Eikon, LEBA, DEHSt

Figure 7: Trading volumes on the primary and secondary market in 2023 and 2024 (EUA)

8 All emission allowances (EUA) of the third trading period traded on the spot and futures markets on the ICE, EEX and Nasdaq OMX exchanges are considered. Transactions completed directly on the stock exchange (screen traded) or those bilateral transactions that at least used the stock exchange clearing mechanism are included. The EUA spot contracts auctioned in 2024 on the EEX are also considered. As far as data on OTC volumes from bilateral trading activity (i. e. off-exchange) is available, it is included in the analysis. Option contracts are not involved.

9 LEBA is an association of major brokerage firms involved in trading on the energy markets. Their monthly volume reports also include bilaterally settled contracts that were not registered on clearing platforms ("OMP Bilaterally Settled Volume"). The data are available as monthly aggregated values and are not broken down into futures-/spot-contracts.

The secondary market trading volume of about 11.2 billion EUA (2023: 8.7 billion) was divided into approximately 10.2 billion EUA (2023: 7.9 billion) on the cleared futures market, some 920 million EUA (2023: 840 million) on the cleared spot market and about 31 million EUA (2023: 12 million) bilaterally settled through LEBA brokers. This represents a significant increase in secondary market trading volumes across all segments, with market shares shown in the figure below.

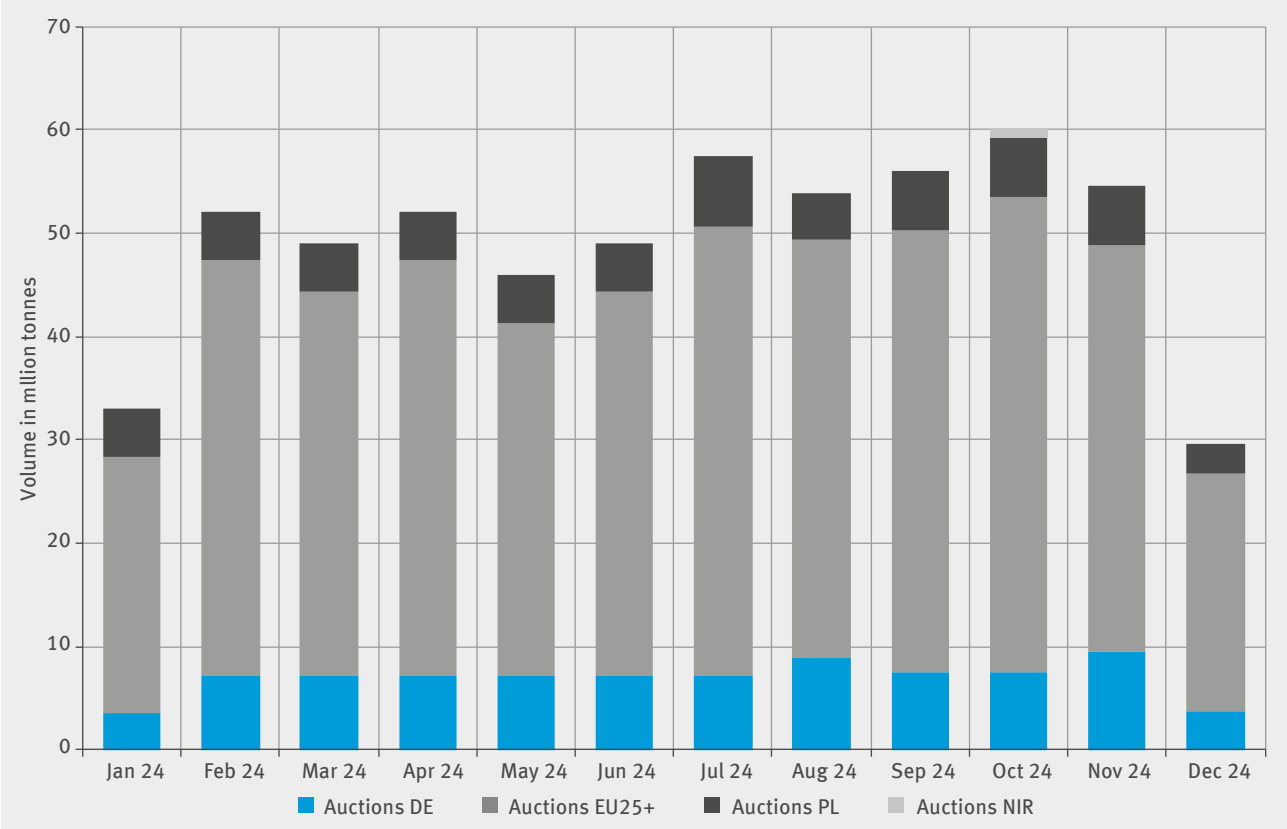


Source: ICE, EEX, Nasdaq OMX, Refinitiv Eikon, LEBA, DEHSt

Figure 8: Market shares of the various trading segments in 2023 and 2024 (EUA)

In 2024, a total of about 593 million allowances (2023: 518 million) were auctioned by Member States on the primary market as spot contracts. Thereof, about 14 percent of the EU-wide auction volume was auctioned by Germany. The increase in auction volumes compared to the previous year is mainly due to the lower reduction by the MSR and the integration of the maritime transport sector into the EU ETS 1. The EU-wide auction volume was relatively evenly distributed over the entire year. However in August, the auction volumes were halved compared to the other months as required by the EU Auctioning Regulation. Only for the first weeks of the year as well as for the two weeks over Christmas and New Year auctioning of allowances rested completely in accordance with European requirements. Based on the total relevant market for emission allowances, about 5 percent of the volume traded can be attributed to the auctions (2023: 5.6 percent).

The following figure shows the auction volumes on the primary market per calendar week (CW). In addition to the German auctions, auction volumes of other EU Member States, as well as Norway, Iceland, Liechtenstein, Northern Ireland[10], for the EU Innovation/Modernisation funds and Recovery and Resilience Facility (RRF; from 2023) are also included in the figure.



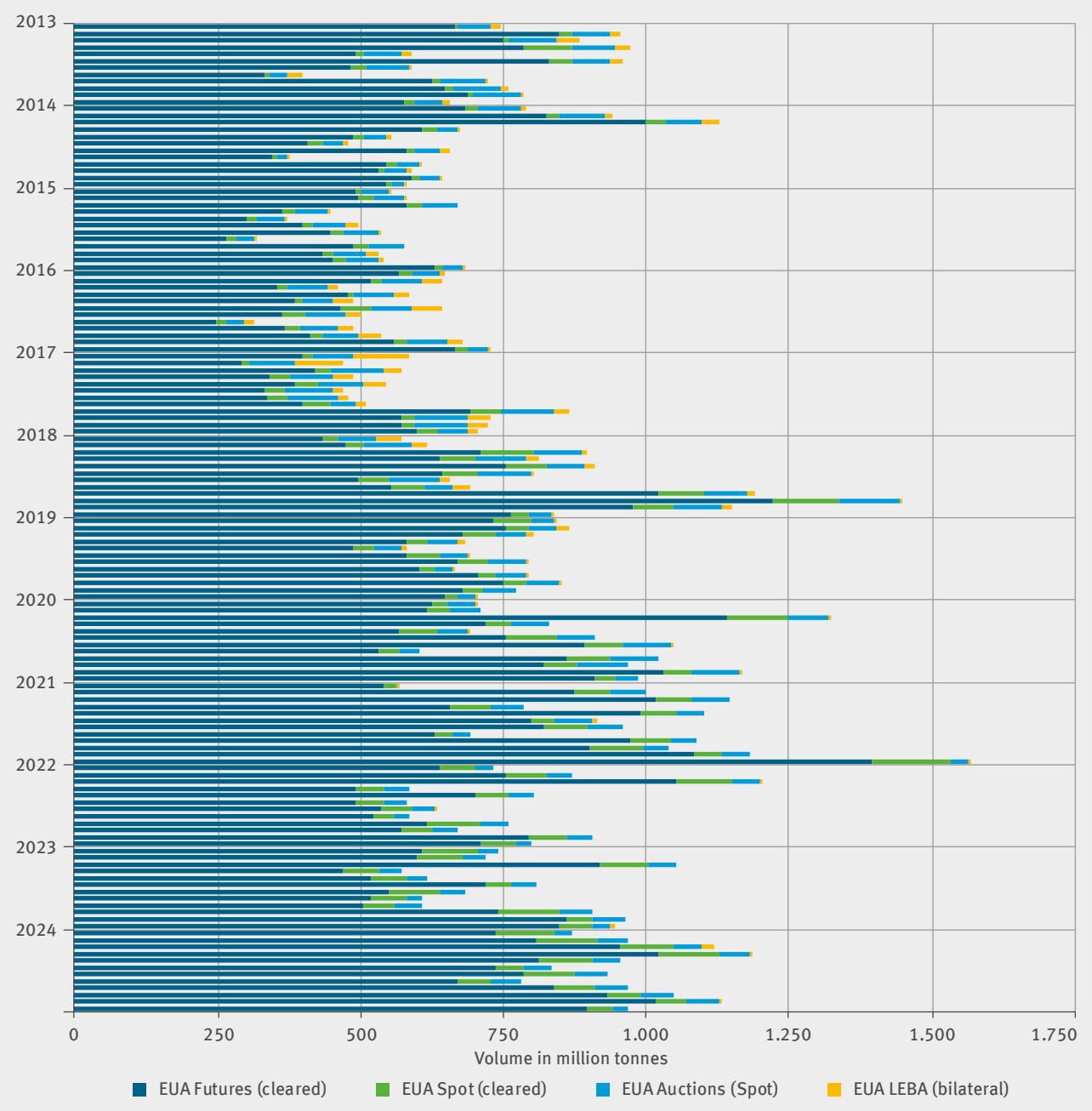
Source: ICE, EEX, Refinitiv Eikon, DEHSt
DE = Germany, PL = Poland, EU25+ = EU-Commission on behalf of other member states and Norway, Iceland and Liechtenstein, NIR = Northern Ireland; EU Innovation/Modernisation funds, Recovery and Resilience Facility (RRF)

Figure 9: Auction volumes on the primary market in 2024 (EUA)

10 In 2021, the UK left the EU. However, pursuant to the Protocol of Ireland and Northern Ireland, the EU ETS continues to apply to generation of electricity in Northern Ireland. Therefore, a small volume of allowances will be auctioned on the EEX on behalf of the UK, corresponding to its share in the Union-wide cap.

3.2 Trading Volume EUA since 2013

The following chart shows the monthly EUA trading volumes since the beginning of the third trading period. In December 2021, the highest trading activity on a monthly basis was observed with a total of around 1.57 billion allowances. Since 2013, the one billion EUA mark was exceeded 21 times, including 7 times in 2021, only once in each of the years 2022 and 2023 and 4 times in 2024. The lowest volumes were achieved in the years 2013 to 2016 with less than 400 million allowances during the "holiday month" of August, when auction volumes were halved in accordance with the requirements of the EU Auctioning Regulation in force at the time. In the past years, however, a clear correlation between auction amounts and trading volumes in August could not always be observed. From 2024 onwards, the volumes to be auctioned in August are no longer reduced by half as in the past, this is in line with the requirements of the new EU Auctioning Regulation.



Source: ICE, EEX, CME, Nasdaq OMX, Refinitiv Eikon, LEBA, DEHSt

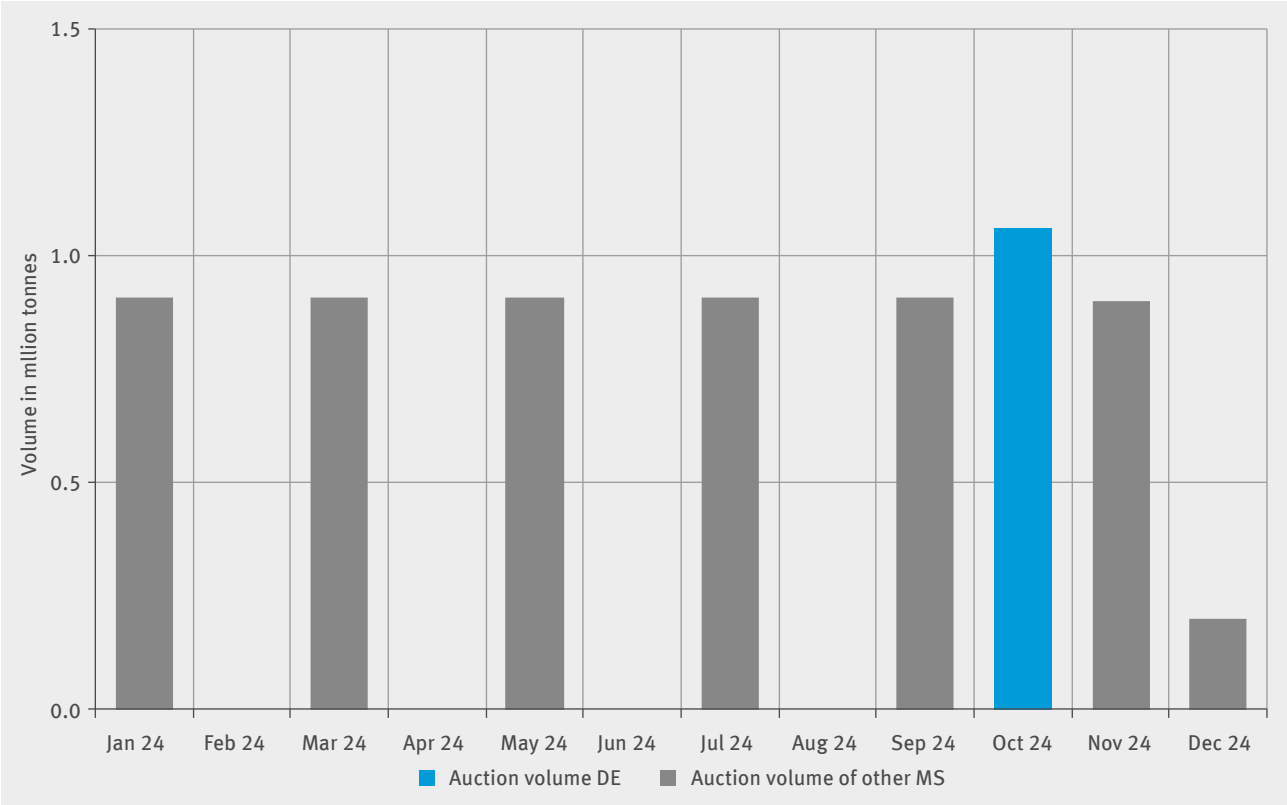
Figure 10: Trading volumes on the primary and secondary market since 2013 (EUA)

3.3 Trading Volume EUAA

In 2024, as in the previous year, no aviation allowances (EUAA) were traded on the secondary market[11]. Liquidity has also remained very low compared to the secondary EUA market in recent years. During the reporting period, a total of about 6.7 million EUAA (2023: 5.7 million) were auctioned by Member States on the primary market as spot contracts.

From 2025 onwards, aviation allowances (EUAA) will no longer be auctioned on the primary market. The corresponding volumes will be auctioned proportionally as EUA spot contracts in the weekly auctions. This is in line with the requirements of the new EU Auctioning Regulation. Further information on the distribution of EUAA auction volumes between calendar years is available on the website of the European Commission.

The figure below shows the monthly auction volumes on the primary market.



Source: EEX, DEHSt
DE = Germany, MS = EU Member States

Figure 11: Auction volumes on the primary market in 2024 (EUAA)

3.4 Auction Prices and Price Development on the Secondary Market

Reference prices from the most liquid trading venues served as appropriate benchmarks for evaluating the auction clearing prices achieved on the EUA spot auctions. The reference contract for the auctions is the EUA daily future traded on the ICE Endex or the EEX EUA spot contract. The product specifications are comparable with the spot-product of the auctions. The trade price immediately before the end of the bidding period was used as a reference price. If no trades take place immediately before the end of the bidding period, the last best bid is used as a reference price (see also Info Box below).

11 All aviation allowances (EUAA) traded on the spot and futures markets on the EEX are considered. Transactions completed directly on the stock exchange (screen traded) or those bilateral transactions that at least used the stock exchange clearing mechanism are included. By the end of 2022, ICE has ceased trading EUAA contracts.

Reference Prices

In regular exchange trading (secondary trading), participants enter their buying and selling requests (bids and asks) into an order book which is constantly updated. As in auctions, such buying bids in secondary trading include quantity and price limits. The price limit is the highest price a buyer is prepared to pay for the purchase of an allowance. Sellers can also put a price limit on their bid, i. e. a minimum price per allowance sold.

The order book is open to inspection by any participant. It shows the current highest purchase offer (best bid) and the lowest sales offer (best ask) with the relevant quantities. Trading occurs where best bid and best ask overlap. Trading price records finally show the price limits where supply equals demand.

Table 3 shows the relative deviations of the clearing price from the EUA lead markets based on monthly averages. For a mean value analysis, the arithmetic mean of the absolute values of individual deviations per auction date is calculated. This prevents positive and negative deviations balancing each other when averaged, thus, the resulting values correspond to the average absolute deviations per month.

The monthly average deviation from the ICE daily future/EEX spot observed immediately before the end of the bidding period (11:00 am), varied during the reporting period between € 0.09 and € 0.98 (2023: between € 0.15 and € 0.60). The relative monthly deviations ranged between 0.14 and 1.60 percent (2023: between 0.18 and 0.69 percent). The average deviation over the full year was € 0.23 or 0.36 percent (2023: € 0.29 or 0.34 percent).

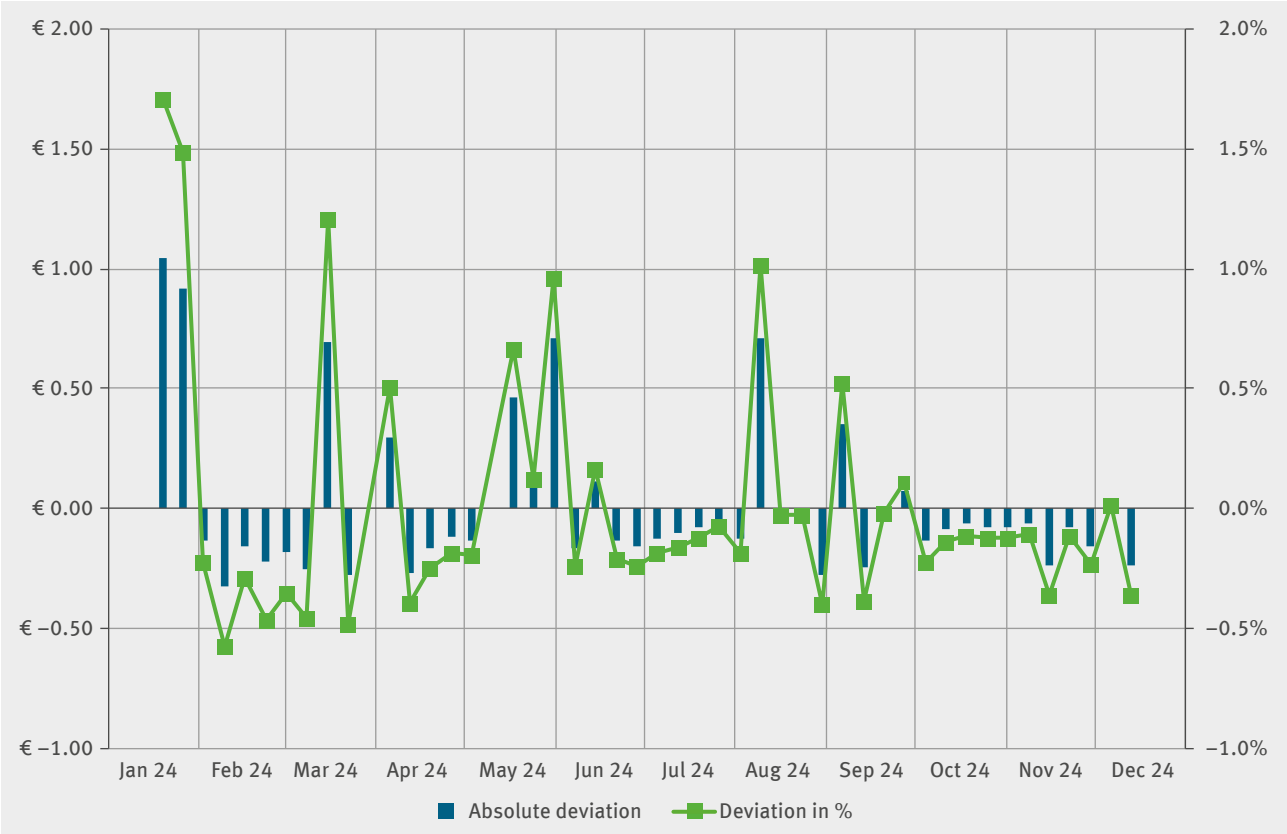
Table 3: EUA auctions and deviations from secondary market prices in 2024

Date	Contract	Clearing price	Deviation from secondary market***	
			Absolute*	%*
January	EUA Spot 4th TP	*€ 61.94	€ 0.98	1.60%
February	EUA Spot 4th TP	*€ 55.70	€ 0.22	0.39%
March	EUA Spot 4th TP	*€ 56.60	€ 0.35	0.62%
April	EUA Spot 4th TP	*€ 64.65	€ 0.21	0.33%
May	EUA Spot 4th TP	*€ 71.98	€ 0.35	0.49%
June	EUA Spot 4th TP	*€ 67.67	€ 0.14	0.21%
July	EUA Spot 4th TP	*€ 66.70	€ 0.09	0.14%
August	EUA Spot 4th TP	**€ 70.40	€ 0.23	0.33%
September	EUA Spot 4th TP	*€ 65.13	€ 0.17	0.26%
October	EUA Spot 4th TP	*€ 63.24	€ 0.10	0.15%
November	EUA Spot 4th TP	*€ 66.89	€ 0.13	0.19%
December	EUA Spot 4th TP	**€ 66.66	€ 0.13	0.19%
Mean absolute deviation			**€ 0.23**	**0.36%**

Source: ICE, EEX, Refinitiv Eikon, DEHSt
* Average
** Volume-weighted average
*** ICE EUA Daily Future/EEX EUA Spot

In addition to the table above, the following figure illustrates the actual deviations of the clearing prices from the secondary spot market for each of the 46 EUA auction dates. A statistical arithmetic mean assessment has not been carried out here, so the difference between positive and negative deviations can be seen. Clearing prices that were below the respective last trade or best bid, lead to a negative deviation. However, if the clearing prices were above the reference price in the secondary market, the figure shows a positive deviation.

Over the year, deviations fluctuated within the range of minus € 0.33 and plus € 1.04 (blue bars). Depending on the particular price level, this resulted in a relative deviation interval of minus 0.57 percent to plus 1.71 percent (green line). In a total of 12 auctions, the auction price was above the secondary market price and in 34 auctions it was below. The relative deviation averaged over all 46 dates was only around plus 0.01 percent (plus € 0.01). Positive and negative deviations have thus almost completely balanced each other out throughout the year.



Source: ICE, EEX, Refinitiv Eikon, DEHSt

Figure 12: Deviations of the clearing price from secondary market price in 2024

The following figure shows an overview of the price development for emissions allowances on the Intercontinental Exchange (ICE Endex), the lead market for EUAs. In addition to the daily futures the development of futures for December delivery of the current and the following year is also shown ("front-December future")[12]. Rather than following the common practice of just looking at settlement prices, our analysis takes into account all trades completed at the exchanges during the reporting period.[13]

12 ICE EUA Dec24 contract expired on 16/12/2024.

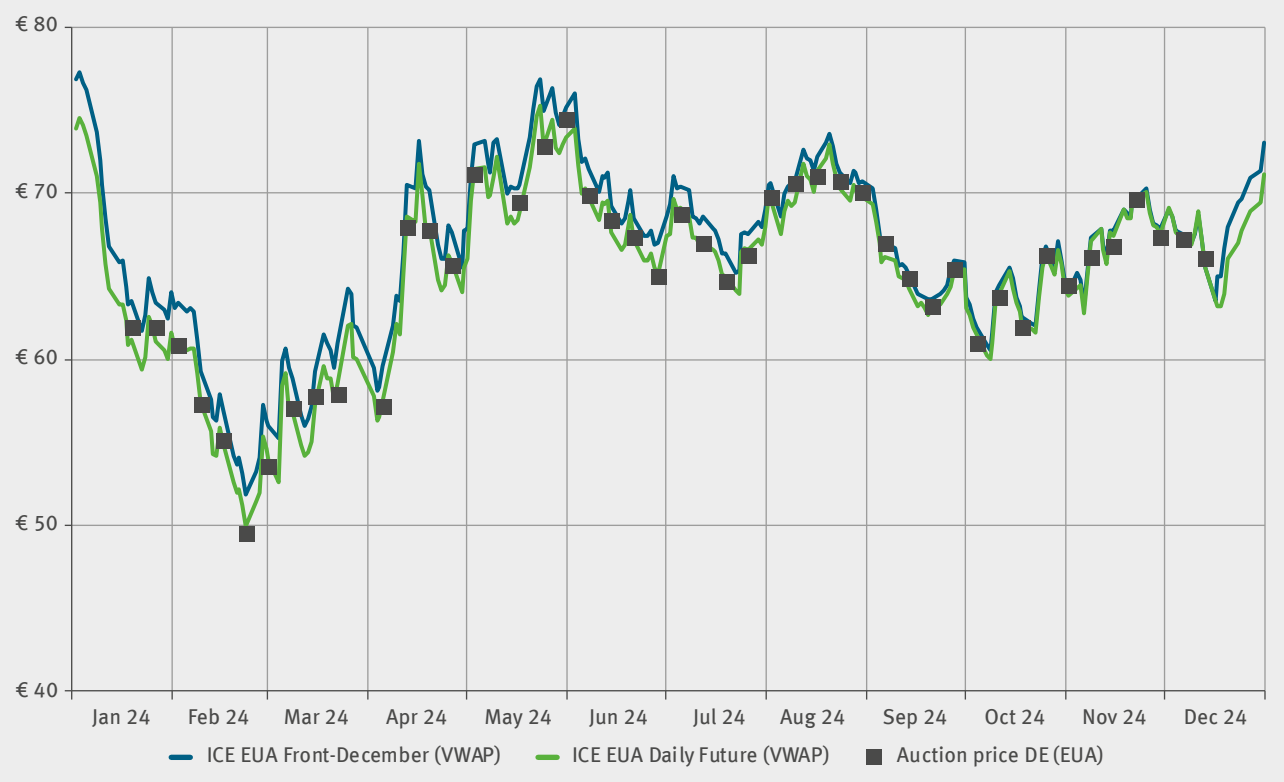
13 Settlement prices are determined by the exchanges at the end of the trading day. Hence, they only reflect trade activity immediately before closure of the exchange and are, thus, not always an adequate reflection of the prevailing intra-day prices in the secondary market.

The following graphic is based on volume-weighted daily average trading prices (VWAP). In 2024, these volume-weighted daily average prices (here front-December Future) fluctuated in a range between € 51.82 and € 77.26 (2023: € 66.92 to € 99.78). At € 66.57, the average value was around 22 percent lower than the previous year's value (2023: € 85.29).[14] The trading range in the trading course of the front-December futures contract in 2024 was between € 51,08 and € 81.25 (2023: € 65,99 to € 101,25).

After a strong upward trend in 2021, when the EUA price gained almost 150 percent in value, and a following sideways movement with high price volatility in 2022 and 2023, the year 2024 was also characterised by strong price fluctuations, especially in the first half of the year (see also Figure 14).

The year 2024 began with a significant downward movement. In February 2024, the EUA price approached the € 50 mark, reaching its lowest level since July 2021, before recovering and rising to over € 70 by May. In the second half of the year, the EUA price moved sideways. At the end of December 2024, the reference price stood at € 73, around 9 percent below the value at the beginning of the year. This was the second annual decline in a row.

In addition to the secondary market prices, the following figure also shows the German auction clearing prices. In principle, the auction clearing prices followed the prevailing prices on the secondary market. The relevant deviations between auction clearing price and the secondary market price per auction date are shown in Figure 12.[15]



Source: ICE, EEX, Refinitiv Eikon, DEHSt

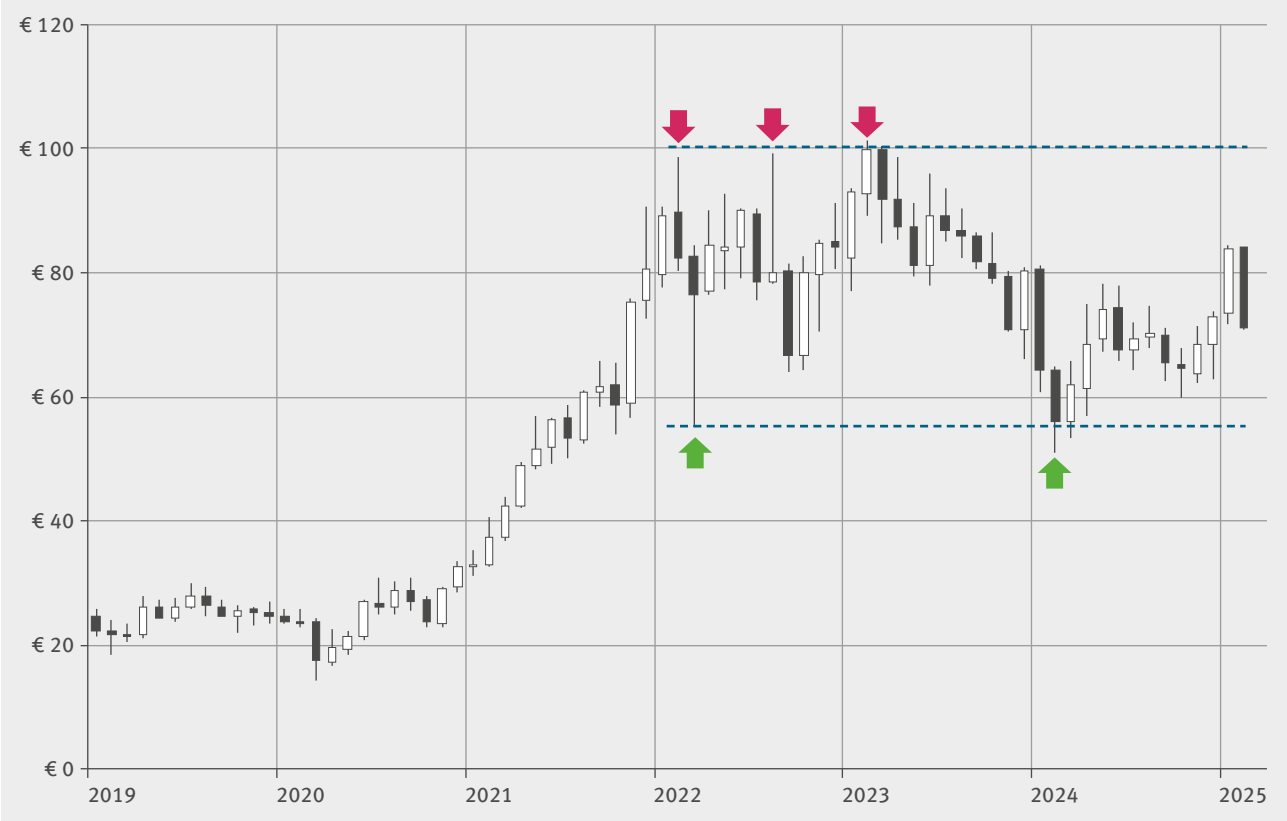
Figure 13: ICE secondary market prices and EEX auction clearing prices (EUA) in 2024

14 The volume-weighted average prices (VWAP) for the ICE EUA front-December future and the ICE daily future contract is based on data from Refinitiv Eikon.

15 It should be noted that the VWAP takes into account all trades completed at the exchanges during the full trading day (green, blue lines), while the auctioning price only reflects the trade activity at a particular time of the day (black dots). Deviations between the two variables are therefore usually attributed to price movements within the trading day.

The following figure also shows the secondary market prices on the ICE as monthly candlesticks since 2019. This illustrates not only the settlement prices for a given period, but also the fluctuation range in addition to the opening and closing prices. After a significant increase in 2021, the EUA reference contract has initially tended to move sideways in a range of around € 55 and € 100 since 2022 (blue lines). The € 100 mark has since been tested three times without success (February and August 2022, February/March 2023). The € 55 mark has been maintained twice to date (March 2022, February/March 2024).

The first two months of 2025 were characterised by contrasting price movements. After reaching its highest level since October 2023 in January at just under € 85, the EUA price moved back towards the € 70 mark in February.



Source: ICE, Refinitiv Eikon, DEHSt (As of: 28/02/2025);
Monthly Candlesticks, ICE EUA front-December Future

Figure 14: ICE secondary market prices since 2019 (EUA)

4 Implementation of Valid Regulations

4.1 Surveillance Report by HÜSt

In order to guarantee above-board market activities are free of manipulation on the EEX continuous surveillance is conducted by an independent Market Surveillance Office (Handelsüberwachungsstelle, HÜSt) as required under German law. HÜSt reports to the DEHSt on the results of its surveillance activities through regular internal reports. HÜSt reported on auctions in 2024 as follows (excerpt):

"Within their surveillance for the auctioning of emission allowances on behalf of the German Environment Agency at the EEX in 2024 the Market Surveillance Office of the EEX could not detect any irregularities and especially no behaviour of bidders that was targeted at manipulating the auction price."

4.2 Fair and Open Access

The implementation of the provisions of the EU Auctioning Regulation guarantees a fair and open access for all auction participants at EEX. Current access options can be found at EEX Website (Access to the auctions).

More information about the auction process is also available on the EEX auction hotline. EEX can be reached by phone from Mondays to Fridays from 08:00 am to 18:00 pm at +49 341 2156-331 or by e-mail at auctions@eex.com.

4.3 Auction Result Publication

All auction results and accompanying data were made publicly available during the reporting period within a 15-minute time window directly linked to the respective auction at EEX Website.

4.4 Other Events

There are no matters relating to the implementation of the existing rules to be reported during the reporting period. The current regulations can be downloaded from EEX Website (Rules and Regulations).

Further information:

www.dehst.de/Auctioning

www.dehst.de/Auctioning-reports